

OAS-010 — Event Architecture

Status: Foundational

Authority: Architectural

Depends on:

- OAS-000
 - OAS-001
 - OAS-002
 - OAS-003
 - OAS-004
 - OAS-005
 - OAS-006
 - OAS-007
 - OAS-008
 - OAS-009
 - OCS-000 through OCS-032
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Purpose

This specification defines the architectural role of Events.

Events are the architectural mechanism through which constitutional history becomes observable.

Architecture SHALL organize around Events rather than mutable state.

State is a temporary interpretation.

Events are permanent constitutional history.

Every implementation SHALL preserve Events as the architectural source of continuity.

1. Architectural Objective

The objective of Event Architecture is to ensure that every constitutional change remains permanently reconstructable.

Architecture SHALL preserve:

What happened.

Why it happened.

Who participated.

What changed.

What remained unchanged.

History SHALL become the architectural foundation.

2. Definition

An Event is an immutable architectural record describing an observable constitutional change.

An Event SHALL possess:

Identity.

Time.

Origin.

Context.

Relationships.

Historical permanence.

Events SHALL remain independently referenceable.

3. Immutability

Events SHALL be immutable.

Corrections SHALL produce new Events.

Interpretations MAY evolve.

Historical Events SHALL remain preserved.

Architecture SHALL never rewrite constitutional history.

4. Architectural State

Architectural state SHALL be understood as a derived interpretation of historical Events.

State MAY be regenerated.

Events SHALL remain primary.

Loss of state SHALL NOT imply loss of constitutional history.

5. Event Relationships

Events SHALL expose explicit relationships.

Relationships MAY include:

Precedes.

Caused.

References.

Supersedes.

Corrects.

Depends Upon.

Relationships SHALL remain reconstructable.

6. Event Composition

Complex constitutional behavior MAY emerge from multiple Events.

Composition SHALL preserve:

Individual identity.

Historical ordering.

Relationship integrity.

Constitutional meaning.

Composite behavior SHALL remain explainable.

7. Event Streams

Every constitutional construct SHALL possess an observable Event Stream.

Event Streams SHALL preserve:

Creation.

Revision.

Participation.

Migration.

Federation.

Historical evolution.

Streams SHALL remain independently reconstructable.

8. Event Time

Architectural time SHALL preserve ordering.

Ordering SHALL remain independent of physical storage.

Equivalent implementations SHALL preserve constitutional sequence.

Exact implementation timestamps SHALL remain implementation concerns.

9. Event Recovery

Architectural recovery SHALL reconstruct constitutional behavior using Event history.

Recovery SHALL preserve:

Identity.

Meaning.

Relationships.

Context.

Historical continuity.

Events SHALL enable complete reconstruction.

10. Event Federation

Events MAY participate in Federation.

Federation SHALL preserve:

Origin.

Ordering.

Context.

Historical attribution.

Federated Events SHALL remain locally owned.

11. Event Verification

Verification SHALL demonstrate:

Historical completeness.

Identity continuity.

Relationship preservation.

Reconstructability.

Equivalent constitutional history.

Implementation SHALL remain replaceable.

12. Event Pattern

Architecture SHALL prefer Event-oriented organization over mutable object-oriented organization whenever constitutional continuity is affected.

Objects represent current interpretation.

Events preserve constitutional memory.

13. Event Architecture Test

Event Architecture satisfies this specification when independent architects can reconstruct identical constitutional history from independent implementations despite differences in storage, execution, communication, and technology.

Equivalent Event history SHALL determine correctness.

Conformance

An implementation conforms to this specification when every constitutionally significant change is represented as an immutable Event whose identity, history, relationships, and meaning remain reconstructable independently of implementation technology.

Architectural Principle

Systems often remember what they are.

Organodynamics remembers how they became what they are.

State describes the present.

Events preserve the journey.

Architecture therefore builds upon history,

because only history allows future generations to understand not merely what exists,

but why it exists.

End of Specification OAS-010